

8. Three integers k , w , and p with $k < w < p$ form a geometric sequence. The sum of these three numbers is 21 and the sum of the reciprocals of these three numbers is $\frac{7}{12}$. Determine these three numbers. Express your answer as the ordered triple (k, w, p) .
9. Andrew and Brent will take turns rolling two fair standard dice until one of two things happens: Andrew gets a sum of 6 or Brent gets a sum of 5. Determine the probability that Brent wins when Brent goes first. Express your answer as an integer or as a common fraction.
10. The period of the graph of $y = 4\sin x \cos x (\cos^2 x - \sin^2 x)$ can be expressed as $k\pi$. Determine the value of k . Express your answer as an integer or as a common or improper fraction.
11. The equation $y = 10e^{(-t)} \cos\left(\frac{t}{3}\right)$ represents a damped vibration, such as the displacement from equilibrium of the end of a stretched spring or the end of struck tuning fork. Determine the smallest value for t after which $|y|$ never exceeds ten per cent of its maximum value. Express your answer as a decimal rounded to the nearest thousandth.
12. $f(n+1) = \frac{3f(n)+1}{3}$ and $f(1) = 5$. Determine the value of $f(2019)$. Express your answer as an integer or as a common or improper fraction.
13. Aune and Ella are trying to meet at the library. Aune can arrive between 3 pm and 5 pm and stay for 17 minutes. Ella can arrive between 3:30 pm and 4:30 pm and stay for 34 minutes. Determine the probability they will be at the library at the same time. Express your answer as an integer or as a common fraction.
14. Let a be a positive integer chosen at random from the interval $[1, 100]$. Then let $k = \frac{a}{2} + \frac{a}{3} + \frac{a}{4} + \frac{a}{5} + \frac{a}{6} + \frac{a}{7}$. Determine the probability that when k is expressed as a simplified and reduced fraction, the numerator of k will be prime. Express your answer as an integer or as a common fraction.